

Module 2 – Algorithms – Data Structures

```
1. import java.util.ArrayList;

import java.util.HashMap;
import java.util.HashSet;
import java.util.List;
import java.util.Map;
import java.util.Set;

public class EcommerceSearch {
    private final List<Product> products;
    private final Map<String, List<Integer>> index = new HashMap<>();

    public static class Product {
        public final String title;
        public final String description;

        public Product(String title, String description) {
            this.title = title;
            this.description = description;
        }

        @Override
        public String toString() {
            return "Product{" + "title='" + title + '\'' + ", description='" +
description + '\'' + '}';
        }
    }

    public EcommerceSearch(List<Product> products) {
        this.products = products;
        buildIndex();
    }

    private void buildIndex(){
        for (int i = 0; i < products.size(); i++) {
            String combined = (products.get(i).title + " " +
products.get(i).description).toLowerCase();
            for (String token : combined.split("\\s+")) {
                index.computeIfAbsent(token, k -> new ArrayList<>()).add(i);
            }
        }
    }

    public List<Product> search(String query) {
        String[] tokens = query.toLowerCase().split("\\s+");
        if (tokens.length == 0) {
            return List.of();
        }
    }
}
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    }
    Set<Integer> matched = null;
    for (String token : tokens) {
        List<Integer> ids = index.getDefault(token, List.of());
        if (matched == null) {
            matched = new HashSet<>(ids);
        } else {
            matched.retainAll(ids);
        }
    }
    if (matched == null) {
        return List.of();
    }
    List<Product> results = new ArrayList<>();
    matched.stream().sorted().forEach(i -> results.add(products.get(i)));
    return results;
}

public static void main(String[] args) {
    List<Product> products = List.of(
        new Product("Red Sneakers", "Comfortable running shoes"),
        new Product("Blue Jeans", "Slim fit denim"),
        new Product("Running Shirt", "Breathable workout top")
    );
    EcommerceSearch engine = new EcommerceSearch(products);
    System.out.println(engine.search("running"));
    System.out.println(engine.search("blue"));
}
}

```

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PS C:\Users\Dedeepya Ramidi\OneDrive\Documents\DSA-2> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-d
sers\Dedeepya Ramidi\AppData\Roaming\Code\User\workspaceStorage\9a31f3a11a24192592242d767720b504\redhat.java\jdt_ws\DSA-2_bc1e4048\bin' 'Ecommerce
[Product{title='Red Sneakers', description='Comfortable running shoes'}, Product{title='Running Shirt', description='Breathable workout top'}]
[Product{title='Blue Jeans', description='Slim fit denim'}]

```

```

2. import java.util.ArrayList;

import java.util.List;
import java.util.Optional;

public class EmployeeManagement {
    public static class Employee {
        public final String employeeId;
        public final String name;
        public final String department;

        public Employee(String employeeId, String name, String department) {
            this.employeeId = employeeId;
            this.name = name;

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        this.department = department;
    }

    @Override
    public String toString() {
        return "Employee{" + "employeeId='" + employeeId + '\'' + ", "
            + name + '\'' + ", department='" + department + '\'' + '}';
    }
}

private final List<Employee> employees = new ArrayList<>();

public void addEmployee(String employeeId, String name, String department)
{
    employees.add(new Employee(employeeId, name, department));
}

public Optional<Employee> findById(String employeeId) {
    return employees.stream()
        .filter(e -> e.employeeId.equals(employeeId))
        .findFirst();
}

public boolean removeEmployee(String employeeId) {
    return employees.removeIf(e -> e.employeeId.equals(employeeId));
}

public List<Employee> listEmployees() {
    return new ArrayList<>(employees);
}

public static void main(String[] args) {
    EmployeeManagement manager = new EmployeeManagement();
    manager.addEmployee("E001", "Alice", "Engineering");
    manager.addEmployee("E002", "Bob", "Sales");
    System.out.println(manager.findById("E001"));
    manager.removeEmployee("E002");
    System.out.println(manager.listEmployees());
}
}

```

```

PS C:\Users\Dedeepya Ramidi\OneDrive\Documents\DSA-2> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCode
'C:\Users\Dedeepya Ramidi\AppData\Roaming\Code\User\workspaceStorage\9a31f3a11a24192592242d767720b504\redhat.java
yeManagement'
Optional[Employee{employeeId='E001', name='Alice', department='Engineering'}]
[Employee{employeeId='E001', name='Alice', department='Engineering'}]

```

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3. import java.util.ArrayList;

import java.util.List;

public class FinancialForecasting {
    public static List<Double> forecastNext(List<Double> values, int steps) {
        if (values.size() < 2) {
            List<Double> result = new ArrayList<>(values);
            for (int i = 0; i < steps; i++) {
                result.add(0.0);
            }
            return result;
        }
        List<Double> forecasted = new ArrayList<>(values);
        for (int i = 0; i < steps; i++) {
            double nextValue = predictNext(forecasted);
            forecasted.add(nextValue);
        }
        return forecasted.subList(values.size(), forecasted.size());
    }

    private static double predictNext(List<Double> values) {
        int size = values.size();
        if (size < 2) {
            return size == 0 ? 0.0 : values.get(0);
        }
        return values.get(size - 1) + (values.get(size - 1) - values.get(size
- 2));
    }

    public static void main(String[] args) {
        List<Double> history = List.of(100.0, 105.0, 110.0);
        System.out.println(forecastNext(history, 3));
    }
}

```

```

PS C:\Users\Dedeepya Ramidi\OneDrive\Documents\DSA-2> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Dedeepya Ramidi\AppData\Roaming\Code\User\workspaceStorage\9a31f3a11a24192592242d767720b504\redhat\va\jdt_ws\DSA-2_bc1e4048\bin' 'FinancialForecasting'
[115.0, 120.0, 125.0]

```

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4. import java.util.ArrayList;

import java.util.HashMap;
import java.util.List;
import java.util.Map;

public class InventoryManagement {
    private final Map<String, Item> inventory = new HashMap<>();
}

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public static class Item {
    public final String sku;
    public final String name;
    public int quantity;
    public final String location;
    public Item(String sku, String name, int quantity, String location) {
        this.sku = sku;
        this.name = name;
        this.quantity = quantity;
        this.location = location;
    }
    @Override
    public String toString() {
        return "Item{" + "sku='" + sku + '\'' + ", name='" + name + '\'' +
", quantity=" + quantity + ", location='" + location + '\'' + '}';
    }
}
public void addItem(String sku, String name, int quantity, String
location) {
    inventory.put(sku, new Item(sku, name, quantity, location));
}
public boolean updateQuantity(String sku, int delta) {
    Item item = inventory.get(sku);
    if (item == null) {
        return false;
    }
    item.quantity += delta;
    return true;
}

public Item getItem(String sku) {
    return inventory.get(sku);
}
public List<Item> listInventory() {
    return new ArrayList<>(inventory.values());
}
public static void main(String[] args) {
    InventoryManagement manager = new InventoryManagement();
    manager.addItem("SKU001", "Widget A", 120, "A1");
    manager.addItem("SKU002", "Widget B", 80, "B2");
    manager.updateQuantity("SKU001", -10);
    System.out.println(manager.getItem("SKU001"));
    System.out.println(manager.listInventory());
}
}

```

```

PS C:\Users\Dedeepya Ramidi\OneDrive\Documents\DSA-2> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInException
\
User\workspaceStorage\9a31f3a11a24192592242d767720b504\redhat.java\jdt_ws\DSA-2_bc1e4048\bin' 'InventoryManagement'
Item{sku='SKU001', name='Widget A', quantity=110, location='A1'}
[Item{sku='SKU001', name='Widget A', quantity=110, location='A1'}, Item{sku='SKU002', name='Widget B', quantity=80, location='B2'}]

```

```

5 import java.util.ArrayList;

import java.util.List;
import java.util.stream.Collectors;

public class LibraryManagement {
    public static class Book {
        public final String isbn;
        public final String title;
        public final String author;

        public Book(String isbn, String title, String author) {
            this.isbn = isbn;
            this.title = title;
            this.author = author;
        }

        @Override
        public String toString() {
            return "Book{" + "isbn='" + isbn + '\'' + ", title='" + title +
            '\'' + ", author='" + author + '\'' + '}';
        }
    }

    private final List<Book> books = new ArrayList<>();

    public void addBook(String isbn, String title, String author) {
        books.add(new Book(isbn, title, author));
    }

    public List<Book> search(String query) {
        String lowered = query.toLowerCase();
        return books.stream()
            .filter(book -> book.title.toLowerCase().contains(lowered)
                || book.author.toLowerCase().contains(lowered))
            .collect(Collectors.toList());
    }

    public static void main(String[] args) {
        LibraryManagement library = new LibraryManagement();
        library.addBook("9780143126560", "The Martian", "Andy Weir");
        library.addBook("9780307277671", "The Road", "Cormac McCarthy");
        System.out.println(library.search("martian"));
        System.out.println(library.search("Cormac"));
    }
}

```

```

PS C:\Users\Dedeepya Ramidi\OneDrive\Documents\DSA-2> & 'C:\Program Files\Java\jdk-24\bin\
epya Ramidi\AppData\Roaming\Code\User\workspaceStorage\9a31f3a11a24192592242d767720b504\re
[Book{isbn='9780143126560', title='The Martian', author='Andy Weir'}]
[Book{isbn='9780307277671', title='The Road', author='Cormac McCarthy'}]

```

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6. import java.util.Comparator;

import java.util.List;
import java.util.stream.Collectors;

public class SortingOrders {
    public static class Order {
        public final String orderId;
        public final double totalPrice;

        public Order(String orderId, double totalPrice) {
            this.orderId = orderId;
            this.totalPrice = totalPrice;
        }

        @Override
        public String toString() {
            return "Order{" + "orderId='" + orderId + '\'' + ", totalPrice=" +
totalPrice + '\'';
        }
    }

    public static List<Order> sortByTotalPrice(List<Order> orders, boolean
descending) {
        return orders.stream()
            .sorted(descending ? Comparator.comparingDouble((Order o) ->
o.totalPrice).reversed()
                : Comparator.comparingDouble(o ->
o.totalPrice))
            .collect(Collectors.toList());
    }

    public static void main(String[] args) {
        List<Order> orders = List.of(
            new Order("1001", 250.0),
            new Order("1002", 75.0),
            new Order("1003", 500.0)
        );
        System.out.println(sortByTotalPrice(orders, true));
    }
}

```

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● PS C:\Users\Dedeepya Ramidi\OneDrive\Documents\DSA-2> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsIn
epya Ramidi\AppData\Roaming\Code\User\workspaceStorage\9a31f3a11a24192592242d767720b504\redhat.java\jdt_ws\DSA-2_bc1e4048\bi
[Order{orderId='1003', totalPrice=500.0}, Order{orderId='1001', totalPrice=250.0}, Order{orderId='1002', totalPrice=75.0}]

```

```

7. public class TaskManagement {

    public static class TaskNode {
        public final String taskId;
        public final String description;
        public TaskNode next;

        public TaskNode(String taskId, String description) {
            this.taskId = taskId;
            this.description = description;
        }

        @Override
        public String toString() {
            return "TaskNode{" + "taskId='" + taskId + '\'' + ",
description='" + description + '\'' + '}';
        }
    }

    private TaskNode head;

    public void addTask(String taskId, String description) {
        TaskNode newNode = new TaskNode(taskId, description);
        if (head == null) {
            head = newNode;
            return;
        }
        TaskNode current = head;
        while (current.next != null) {
            current = current.next;
        }
        current.next = newNode;
    }

    public boolean removeTask(String taskId) {
        TaskNode current = head;
        TaskNode previous = null;
        while (current != null) {
            if (current.taskId.equals(taskId)) {
                if (previous != null) {
                    previous.next = current.next;
                } else {
                    head = current.next;
                }
                return true;
            }
            previous = current;
            current = current.next;
        }
    }
}

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        return false;
    }

    public String traverseTasks() {
        StringBuilder sb = new StringBuilder();
        TaskNode current = head;
        while (current != null) {
            sb.append(current).append("\n");
            current = current.next;
        }
        return sb.toString();
    }

    public static void main(String[] args) {
        TaskManagement tasks = new TaskManagement();
        tasks.addTask("T1", "Review requirements");
        tasks.addTask("T2", "Implement feature");
        tasks.addTask("T3", "Write tests");
        tasks.removeTask("T2");
        System.out.println(tasks.traverseTasks());
    }
}

```

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● PS C:\Users\Dedeepya Ramidi\OneDrive\Documents\DSA-2> c:; cd 'c:\Users\De
XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Dedeepya Ramidi\A
_ws\DSA-2_bc1e4048\bin' 'TaskManagement'
TaskNode{taskId='T1', description='Review requirements'}
TaskNode{taskId='T3', description='Write tests'}

```